AKIB JAVED, Ph.D.

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Employment History

May 2026 – Present
Consultant on Hydrology and Geoinformatics, C4RE (Centre for Research on Environment and People) Directs geospatial research and technical frameworks for evidence-based hydrological project roadmaps.

- **Coastal Hazard Risk Assessment Specialist, UNDP GEF-LDCF WOIA Project:** Developing the Comprehensive Coastal Hazard Risk Assessment and Management across five project islands for the Kiribati Climate Change and Disaster Risk Management (CCDRM) consultancy.
- **GIS Specialist, World Bank KOIRAP Project:** Preparing the *Kiribati Climate Resilient Urban Policy 2026-2036* by utilizing advanced spatial data science, hazard mapping, and image analysis to integrate climate resilience into regional urban development.

Feb 2026 – Present
Founder & Geospatial Consultant, Maplicon. Established and leading a specialized consultancy firm providing research and consultancy services on remote sensing analysis, Google Earth Engine (GEE) automation, and spatial data science solutions.

Mar 2026 – Apr 2026
Research Coordinator (Part-time), Center for People & Environment (CPE). Coordinates the Research division by translating technical research into practical solutions and policy recommendations.

Feb 2025 – Jan 2026
GIS Analyst, Institute of Water Modelling (IWM). Georeferenced and digitized old cadastral maps for a national land zoning project, improving land management efficiency for government agencies and reducing manual processing time by 20% within my team.

Jun 2025 – Aug 2025
GIS Supervisor (Part-time), Center for People & Environment (CPE). Led a GIS team of 3 analysts, improving project delivery speed by 20% and ensuring 95% accuracy in all spatial data outputs.

Jun 2019 – Jun 2019
Internship, Hi-Target Surveying Instrument Co., Ltd., Guangzhou, China. Participated in a week-long internship program and familiarized with the latest geospatial tools and software.

Jul 2017 – Jul 2017
Visiting Lecturer. Taught GIS and remote sensing courses to 100+ undergraduate students, improving average student geospatial knowledge by 5% through hands-on sessions within a week.

Apr 2017 – Aug 2017
Research Assistant. Analyzed land use patterns for the Adivasi People project, generating key insights that informed community development strategies for 3,000+ beneficiaries.

Jul 2016 – Mar 2017
Research Assistant (Part-time). Supported PhD research by analyzing primary data, accelerating data processing time by 35% and contributing to 5 peer-reviewed publications on environmental science.

Mar 2016 – Jun 2016
Research Assistant (Part-time). Assisted Prof. Dr. Raquib Ahmed (Rajshahi University) in environmental change research, utilizing geospatial analysis to increase data accuracy by 10%.

Education

Academic Degrees

2017 – 2024 **Ph.D.** in **Cartography and GIS** from LIESMARS, Wuhan University, China.

2014 – 2015 **M.S.** in **Geography and Environmental Studies** from University of Rajshahi.

2010 – 2013 **B.Sc.** in **Geography and Environmental Studies** from University of Rajshahi.

Scholarship

2017 **CSC Scholarship**: Awarded by Chinese Scholarship Council for doctoral studies

Research

Academic Service

Journal Peer Review — Remote Sensing of Environment, Advances in Space Research, Journal of Selected Topics in Applied Earth Observations and Remote Sensing, City and Environment Interactions

Conference Review — ISPRS Geospatial Week 2025, Dubai

Research Co-supervision

2025 — Spatio-temporal Mapping of Forest Areas of Bangladesh: A remote sensing base study using Google Earth Engine (GEE) platform.

2024 — Spatiotemporal Analysis of Droughts in Northern Bangladesh using MODIS Time Series (2000-2023).

Spatiotemporal Mapping of Urban Air Pollutants: A Comparative Study of Bangladesh.

2026 — Assessing Three Decades of Forest Dynamics in Bangladesh: Spatiotemporal Mapping and Seasonal Typologies via Earth Observation.

Research Experience

PhD thesis — Researched urban impervious surface extraction across space and time using open-source spatial big data.

B.Sc. project — Investigated urban transformation in Rajshahi City, revealing key patterns of urban sprawl driven by road networks and informing sustainable development strategies for 500K+ residents.

2017 — Conducted intensive land study for the Adivasi People project, collecting data from 300+ households and informing community development strategies for 5K+ beneficiaries.

2013 — Authored a detailed field reports from a socioeconomic survey on Shahriadaha Island, providing critical data that informed local lifestyle and economic activities for 3K+ residents.

2012 — Authored a comprehensive field reports from a socioeconomic survey on Dubla Char Island, delivering key insights of local fisherman.

2011 — Authored a field reports on industrial regions of Kushtia, providing critical data that informed major economic activities for four factories.

2010 — Authored a field reports on Historical Importance of Mohasthangarh and Paharpur.

Research Publications

Journal Articles

1. M. N. Ahmad, X. Yuan, L. Gu, S. Yang, R. W. Aslam, **A. Javed**, and G. D. Bathrellos, "Advancements in deep learning methods for precise weed and rice classification from UAV imagery: A comprehensive review," *Discover Agriculture*, vol. 4, no. 1, p. 8, Jan. 2026, ISSN: 2731-9598. DOI: 10.1007/s44279-025-00469-0

2. M. N. Ahmad, X. Yuan, S. Yang, L. Gu, H. Peng, R. W. Aslam, M. E. Huq, **A. Javed**, and G. D. Bathrellos, "Dtsm: A uav-based framework for temporal rice stress detection and growth analysis," *Smart Agricultural Technology*, p. 102 141, Apr. 2026, ISSN: 2772-3755. DOI: https://doi.org/10.1016/j.atech.2026.102141

3. I. Ara, R. I. Zerin, L. A. Eva, M. K. Hossain, M. H. Mou, **A. Javed**, S. Ali, M. A. Kabir, and M. M. P. Rana, "Comparative hydrogeochemical characterization and environmental controls on groundwater quality, irrigation suitability, and health risk across northern and coastal hydrological settings of bangladesh," *Cleaner Water*, vol. 5, p. 100 194, Mar. 2026, ISSN: 2950-2632. DOI: https://doi.org/10.1016/j.clwat.2025.100194

4. L. A. Eva, I. Ara, M. A. Rahman, M. S. Khan, S. Ali, R. I. Zerin, F. Islam, **A. Javed**, M. M. Hasan, M. K. Hossain, et al., "Physicochemical, heavy metal, and microbial risk assessment of harvested rainwater in coastal koyra upazila, bangladesh," *Journal of Water, Sanitation and Hygiene for Development*, vol. 16, washdev2026208, May 2026. DOI: https://doi.org/10.2166/washdev.2026.208

5. F. Islam, I. Ara, M. S. Khan, **A. Javed**, M. K. Hossain, M. O. H. Khan, M. A. Kabir, and F. Sadique, "Assessment of particulate matter exposure and associated human health risk: Comparison among non-automated, automated and semi-automated rice industry workers," *Journal of Industrial Safety*, Apr. 2026, ISSN: 2950-2764. DOI: https://doi.org/10.1016/j.jinse.2026.04.001

6. M. N. Ahmad, F. Almutlaq, M. E. Huq, F. Islam, **A. Javed**, H. D. Skilodimou, and G. D. Bathrellos, "Enhanced urban impervious surface land use mapping using a novel multi-sensor feature fusion method and remote sensing data," *Environmental Earth Sciences*, vol. 84, no. 9, p. 228, Apr. 2025, ISSN: 1866-6299. DOI: 10.1007/s12665-025-12217-0

7. M. N. Ahmad, H. D. Skilodimou, F. Islam, **A. Javed**, and G. D. Bathrellos, "Drainage network generation for urban pluvial flooding (upf) using generative adversarial networks (gans) and gis data," *Sustainability*, vol. 17, no. 10, May 2025, ISSN: 2071-1050. DOI: 10.3390/su17104380

8. R. I. Zerin, I. Ara, S. Ali, M. K. Hossain, **A. Javed**, S. N. Ratna, and M. A. Kabir, "Geostatistical modeling of soil physicochemical properties and environmental drivers in agro-industrial landuse systems in bangladesh," *Geomatica*, vol. 77, no. 2, p. 100 075, Sep. 2025, ISSN: 1195-1036. DOI: https://doi.org/10.1016/j.geomat.2025.100075

9. M. N. Ahmad, Z. Shao, **A. Javed**, I. Ahmad, F. Islam, H. D. Skilodimou, and G. D. Bathrellos, "Optical-sar data fusion based on simple layer stacking and the xgboost algorithm to extract urban impervious surfaces in global alpha cities," *Remote Sensing*, vol. 16, no. 5, 2024. DOI: 10.3390/rs16050873

10. M. N. Ahmad, Z. Shao, X. Xiao, P. Fu, **A. Javed**, and I. Ara, "A novel ensemble learning approach to extract urban impervious surface based on machine learning algorithms using sar and optical data," *International Journal of Applied Earth Observation and Geoinformation*, vol. 132, p. 104 013, 2024, ISSN: 1569-8432. DOI: https://doi.org/10.1016/j.jag.2024.104013

11. M. E. Huq, X. Wu, **A. Javed**, Y. Dong, B. Li, D. Lopez-Carr, J. Wu, J. Liu, Y. Zhang, F. Zhang, M. Riaz, M. N. I. Sarker, W. Soufan, K. F. Almutairi, and A. Tariq, "Changes monitoring in hongjiannao lake from 1987 to 2023 using google earth engine and analysis of climatic and anthropogenic forces," *Physics and Chemistry of the Earth, Parts A/B/C*, vol. 136, p. 103 756, 2024. DOI: https://doi.org/10.1016/j.pce.2024.103756

12. **A. Javed**, Z. Shao, I. Ara, M. N. Ahmad, E. Huq, N. Saleem, and F. Karim, "Development of soil-suppressed impervious surface area index for automatic urban mapping," *Photogrammetric Engineering & Remote Sensing*, vol. 90, no. 1, pp. 33–43, 2024, ISSN: 0099-1112. DOI: 10.14358/PERS.23-00043R2

13 Z. Shao, M. N. Ahmad, and **A. Javed**, "Comparison of random forest and xgboost classifiers using integrated optical and sar features for mapping urban impervious surface," *Remote Sensing*, vol. 16, no. 4, p. 665, 2024, ISSN: 2072-4292. URL: https://www.mdpi.com/2072-4292/16/4/665

14 M. N. Ahmad, Z. Shao, and **A. Javed**, "Modelling land use/land cover (lulc) change dynamics, future prospects, and its environmental impacts based on geospatial data models and remote sensing data," *Environmental Science and Pollution Research*, vol. 30, no. 12, pp. 32 985–33 001, 2023, ISSN: 1614-7499 (Electronic) 0944-1344 (Linking). DOI: 10.1007/s11356-022-24442-2

15 M. N. Ahmad, Z. Shao, and **A. Javed**, "Mapping impervious surface area increase and urban pluvial flooding using sentinel application platform (snap) and remote sensing data," *Environmental Science and Pollution Research*, vol. 30, no. 60, pp. 125 741–125 758, 2023, ISSN: 1614-7499. DOI: 10.1007/s11356-023-30990-y

16 M. N. Ahmad, Z. Shao, **A. Javed**, F. Islam, H. H. Ahmad, and R. W. Aslam, "The cellular automata approach in dynamic modelling of land use change detection and future simulations based on remote sensing data in lahore pakistan," *Photogrammetric Engineering & Remote Sensing*, vol. 89, no. 1, pp. 47–55, 2023, ISSN: 0099-1112. DOI: 10.14358/pers.22-00102r2

17 M. Y. Ali, M. Shahrier, A. A. Kafy, I. Ara, **A. Javed**, M. A. Fattah, Z. A. Rahaman, and K. Tripura, "Environmental impact assessment of tobacco farming in northern bangladesh," *Heliyon*, vol. 9, no. 3, e14505, 2023, ISSN: 2405-8440 (Print) 2405-8440 (Electronic) 2405-8440 (Linking). DOI: 10.1016/j.heliyon.2023.e14505

18 **A. Javed**, Z. F. Shao, B. Bai, Z. Y. Yu, J. B. Wang, I. Ara, M. E. Huq, M. Y. Ali, N. Saleem, M. N. Ahmad, N. S. Sumari, and Mardia, "Development of normalized soil area index for urban studies using remote sensing data," *Geofizika*, vol. 40, no. 1, pp. 29–49, 2023, ISSN: 0352-3659. DOI: 10.15233/gfz.2023.40.2

19 Z. Shao, M. N. Ahmad, **A. Javed**, F. Islam, Z. Jahangir, and I. Ahmad, "Expansion of urban impervious surfaces in lahore (1993-2022) based on gee and remote sensing data," *Photogrammetric Engineering and Remote Sensing*, vol. 89, no. 8, pp. 479–486, 2023, ISSN: 0099-1112. DOI: 10.14358/pers.23-00001r2

20 M. N. Ahmad, Z. F. Shao, A. Yaseen, M. N. Khalid, and **A. Javed**, "The simulation and prediction of land surface temperature based on scp and ca-ann models using remote sensing data: A case study of lahore," *Photogrammetric Engineering and Remote Sensing*, vol. 88, no. 12, pp. 783–790, 2022, ISSN: 0099-1112. DOI: 10.14358/pers.22-00071r2

21 M. Y. Ali, A.-A. Kafy, Z. A. Rahaman, M. F. Islam, M. R. Rahman, I. Ara, M. R. Akhtar, and **A. Javed**, "Comparative occupational health risk between tobacco and paddy farming people in bangladesh," *SSM-Mental Health*, vol. 2, p. 6, 2022.

22 **A. Javed**, Q. M. Cheng, H. Peng, O. Altan, Y. Li, I. Ara, E. Huq, Y. Ali, and N. Saleem, "Review of spectral indices for urban remote sensing," *Photogrammetric Engineering and Remote Sensing*, vol. 87, no. 7, pp. 513–524, 2021, ISSN: 0099-1112. DOI: 10.14358/Pers.87.7.513

23 M. E. Huq, Q. M. Cheng, O. Altan, A. Z. M. Shoeb, M. A. Hossain, M. N. I. Sarker, N. Saleem, **A. Javed**, X. Long, A. A. Al Dughairi, M. M. P. Rana, A. Al Mamun, and M. M. Rahman, "Assessing vulnerability for inhabitants of dhaka city considering flood-hazard exposure," *Geofizika*, vol. 37, no. 2, pp. 97–130, 2020. DOI: 10.15233/gfz.2020.37.5 wos: 000608071700001. Accessed: Oct. 7, 2025.

24 N. Saleem, M. E. Huq, N. Y. D. Twumasi, **A. Javed**, and A. Sajjad, "Parameters derived from and/or used with digital elevation models (dems) for landslide susceptibility mapping and landslide risk assessment: A review," *ISPRS International Journal of Geo-Information*, vol. 8, no. 12, pp. 545–572, 2019.

Conference Proceedings

1 M. S. Akter, I. Ara, and **A. Javed**, "A comprehensive evaluation of environmental literacy: A case study in kushtia municipality, bangladesh," in *ABSTRACTS BOOK OF FIRST INTERNATIONAL CONFERENCE ON BANGLADESH STUDIES (ICBS-I) 2025*, O. AKGÜL, H. Rahman, and A. M. Hasan, Eds., Presented via online, Istanbul Ticaret University, Türkiye: Istanbul Ticaret University, Türkiye, Apr. 2025, p. 24.

2. M. S. Akter, I. Ara, and **A. Javed**, "A comprehensive evaluation of environmental literacy: A case study in kushtia municipality, bangladesh," in *Conference Proceedings on 1st International Conference on Bangladesh Studies (ICBS-I), 2025*, Hybrid conference held on April 19-20, 2025, Istanbul Ticaret University, Türkiye, Apr. 2025. URL: https://cpsrinfo.org/

3. S. Ali, **A. Javed**, and I. Ara, "Agricultural drought in northern bangladesh: A spatiotemporal analysis using modis time series from 2001 to 2023," in *ABSTRACTS BOOK OF FIRST INTERNATIONAL CONFERENCE ON BANGLADESH STUDIES (ICBS-I) 2025*, O. AKGÜL, H. Rahman, and A. M. Hasan, Eds., Presented via online, Istanbul Ticaret University, Türkiye: Istanbul Ticaret University, Türkiye, Apr. 2025, p. 46.

4. L. A. Eva, I. Ara, M. S. K. Khan, M. A. Rahman, and **A. Javed**, "From resource to risk: Evaluating the rainwater quality and associated health issues in the southern coastal bangladesh," in *Conference Proceedings on 10th International Conference on Water and Flood Management - ICWFM 2025*, vol. 10, Institute of Water and Flood Management, BUET, Feb. 2025, pp. 277–284.

5. F. Islam, I. Ara, M. S. Khan, and **A. Javed**, "Assessment of particulate matter exposure and associated human health risk: Comparison among non-automated, automated and semi-automated rice processing industry workers in bangladesh," in *Proceedings of the 1st International Conference on Geosciences for Achieving Sustainable Development Goals (ICGSDG-2025)*, Faculty of Geosciences, University of Rajshahi, Bangladesh, Nov. 2025, p. 162.

6. H. Rashid, **A. Javed**, and I. Ara, "A Remote Sensing Assessment of Air Pollution in Five Bangladeshi Industrial Cities," in *INTERNATIONAL GLOBAL SUSTAINABILITY AND DEVELOPMENT CONGRESS Full Text Book*, O. Özkan and Ş. T. Azgin, Eds., New York, NY: Liberty Publishing House, Nov. 2025, pp. 873–897, ISBN: 979-8-89695-250-3. URL: iksadkongre.com/_files/ugd/614b1f_c112d6b2a09442749bd2d5579d9cdb53.pdf

7. H. Rashid, **A. Javed**, and I. Ara, "A remote sensing assessment of air pollution in five bangladeshi industrial cities," in *Global Sustainability and Development Congress Abstract Book*, O. Özkan and Ş. T. Azgın, Eds., Kayseri, Türkiye: Liberty Academic Publishers, Oct. 2025, p. 245, ISBN: 979-8-89695-212-1.

8. R. R. Ria, I. Ara, and **A. Javed**, "An analysis of governance framework effectiveness and internal waste management practices in the hospitals of rangpur sadar upazila, bangladesh," in *ABSTRACTS BOOK OF FIRST INTERNATIONAL CONFERENCE ON BANGLADESH STUDIES (ICBS-I) 2025*, O. AKGÜL, H. Rahman, and A. M. Hasan, Eds., Presented via online, Istanbul Ticaret University, Türkiye: Istanbul Ticaret University, Türkiye, Apr. 2025, p. 77.

9. R. R. Ria, I. Ara, and **A. Javed**, "An analysis of governance framework effectiveness and internal waste management practices in the hospitals of rangpur sadar upazila, bangladesh," in *Conference Proceedings on 1st International Conference on Bangladesh Studies (ICBS-I), 2025*, Hybrid conference held on April 19-20, 2025, Istanbul Ticaret University, Türkiye, Apr. 2025. URL: https://cpsrinfo.org/

10. R. I. Zerin, I. Ara, L. Anjum, M. H. Moti, M. K. Hossain, and **A. Javed**, "Index-based assessment of groundwater quality and suitability for irrigation across major agricultural regions of bangladesh," in *Proceedings of the 1st International Conference on Geosciences for Achieving Sustainable Development Goals (ICGSDG-2025)*, Faculty of Geosciences, University of Rajshahi, Bangladesh, Nov. 2025, p. 109.

11. M. Huq, S. Zhenfeng, C. Bowen, N. Saleem, A. A. Aldughairi, D. T. N. Yaw, **A. Javed**, and M. M. Rahman, "Measuring vulnerability to flash flood for urban inhabitants: A case study of dhaka megacity, bangladesh," in *The Fifth International Symposium on Flash Floods in Wadi Systems (ISFF2020)*, 2020, p. 87.

12. M. E. Huq, A. Shoeb, **A. Javed**, Z. Shao, M. A. Hossain, and M. S. Sarven, "Measuring vulnerability for city dwellers exposed to flood hazard: A case study of dhaka city, bangladesh," in *Urban Intelligence and Applications: Proceedings of ICUIA 2019*, Springer International Publishing, 2019, pp. 207–215, ISBN: 3030450988.

13. M. Y. Ali, M. Hassan, M. A. Rahmana, A.-A. Kafy, I. Ara, **A. Javed**, and M. R. Rahman, "Life cycle energy and cost analysis of small scale biogas plant and solar pv system in rural areas of bangladesh," in *2nd International Conference on Energy and Power, ICEP2018, 13–15 December 2018, Sydney, Australia*, vol. 160, Energy Procedia, ELSEVIER, 2018, pp. 277–284.

14. **A. Javed** and S. Zhenfeng, "Smart application using gnss, gis and remote sensing: Flood prediction and monitoring," in *First International Workshop on Artificial Intelligence and Big Data Analysis 2017*, 2017, p. 67.

15. **A. Javed** and R. Ahmed, "Philosophy of geography education: Some perspectives and progress in bangladesh context," in *International Conference on Geography Education 2015*, 2015, p. 33.

Books and Chapters

1. M. E. Huq, D. Lopez-Carr, A. Tariq, X. Wu, J. Wu, Y. Dong, Z. Daud, **A. Javed**, M. N. I. Sarker, K. M. Adnan, et al., "Advancements in remote sensing and gis for water resource management and sdg monitoring," in *Utilizing Earth Observation Data in Reaching Sustainable Development Goals*. Elsevier, 2026, pp. 189–218. DOI: https://doi.org/10.1016/B978-0-443-30204-6.00007-3

2. M. E. Huq, Z. Shao, A. A. Al Dughairi, M. N. I. Sarker, C. Bowen, A. Al Mamun, N. Saleem, **A. Javed**, and M. M. Rahman, "Measuring vulnerability to flash flood of urban dwellers," in *Wadi Flash Floods*. Springer, Singapore, 2022, p. 317.

3. M. E. Huq, A. Z. M. Shoeb, M. A. Hossain, S. Fahad, M. M. Kamruzzaman, **A. Javed**, N. Saleem, K. M. M. Adnan, S. A. Sarker, and M. Y. Ali, "Measuring vulnerability to environmental hazards: Qualitative to quantitative," in *Environment, Climate, Plant and Vegetation Growth*. Springer, 2020, pp. 421–452.

Posters

1. **A. Javed** and M. N. Ahmad, "Chinese desert reclamation: Turning barren land into green oases," Mar. 2023. DOI: 10.13140/RG.2.2.30223.46240

2. **A. Javed**, Z. Shao, I. Ara, M. N. Ahmad, E. Huq, N. Saleem, and F. Karim, "Soil suppressed generalized impervious surface area index (gisai) for automatic megacity mapping," Mar. 2023. DOI: 10.13140/RG.2.2.35676.05761

Seminars, Conferences, Workshops, Trainings

Seminars

2023-10-12 **Mobile Mapping Systems: Demonstrated Technology that Keeps Evolving** by Prof. Ayman F. Habib, at LIESMARS, Wuhan University.

2023-10-03 **Unleashing the Power of Words: Writing Practices and Processes** by Stephen C. McClure, at LIESMARS, Wuhan University.

2023-06-20 **Smart City** by Prof. Dr. Zhenfeng Shao, at LIESMARS, Wuhan University.

2023-05-30 **Navigating the Academic Landscape: Insights and Strategies for Successful Research and Career Development** by Prof. Naser El-Sheimy, at LIESMARS, Wuhan University.

2023-05-21 **Forest Aboveground Biomass Estimation with Airborne LiDAR data** by Prof. Dengsheng Lu, at LIESMARS, Wuhan University.

2023-05-06 **LEO Navigation Augmentation System from Luojia-1A/2A Satellite - Design, Progress and Application** by Prof. Lei Wang, at LIESMARS, Wuhan University.

Conferences

17-19 November 2024 Participated in the **2024 International Symposium on Spatio-Temporal Intelligence and 2024 International Conference on Geospatial Information Science: Education, Innovation and Applications** organized by LIESMARS, Wuhan University.

Seminars, Conferences, Workshops, Trainings (continued)

15-16 October 2023 — Participated in the **International Conference on Geospatial Information Science: Education, Innovation and Applications** organized by LIESMARS, Wuhan University.

16-18 October 2018 — Participated in the **International Conference on Advance Remote Sensing 2018** organized by LIESMARS, Wuhan University.

20 December 2017 — Participated in the **International Symposium on key technologies and applications of Video Surveillance and Intelligent Analysis in Hengqin Island** organized by Zhuhai City Development Authority, Guangzhou, Guangdong, China.

11-13 September 2015 — Participated in the **International Conference on Geography Education 2015** organized by the Dept. of Geography and Environmental Studies, University of Rajshahi, Bangladesh and Institute of Geography, University of Freiburg, Germany at the University of Rajshahi, Bangladesh.

07-08 March 2014 — Participated in the **International Conference on Environment and Development** organized by the Department of Geography and Environmental Studies, University of Rajshahi, supported by Higher Education Quality Enhancement Project (HEQEP).

Workshop

2019 — Ten days long Workshop title **Zhuhai One Hyperspectral Remote Sensing Training** organized by Zhuhai ORBITA Aerospace Science & technology Co., Ltd. at Obit Science and Technology Park, No. 1 Baisha Road, Tangjia East Bank, Zhuhai City, China.

2017 — Three days long workshop title **The 1st LIESMARS International Graduate Workshop on GeoInformatics** organized by LIESMARS, Wuhan University, Wuhan City, China (November 25-27, 2017).

2014 — Participated to a Workshop on **Teaching Method and Curriculum Development** at the Department of Geography and Environmental Studies, University of Rajshahi, Bangladesh: Organized by University of Education Freiburg, Germany and University of Rajshahi, Bangladesh (November 10-12, 2014).

2013 — Participated on Teacher Training Workshop on **Innovative Teaching Methodology, Learning Strategies and Didactics** at the University of Rajshahi, Bangladesh. Organized by the University of Education Freiburg, Germany and the University of Rajshahi, Bangladesh (18-20 February 2013).

Training and Summer School

2024 — Completed ten-day long training Program titled **Application of Mathematical Modelling & GIS Technology for Feasibility Study of Water Development Projects** organized by Institute of Water Modelling (IWM), Dhaka, Bangladesh (29 September - 09 October 2024).

23-29 June, 2024 — Participated in **2024 Luojia Global Summer School** organized by Wuhan University. Took courses on **Social Sensing and Big Data Computing for Disaster Management** with the instructor Prof Dr. Zhenlong Li from The Pennsylvania State University, USA.

12-18 July, 2023 — Participated in **2023 International Geo-Informatics Summer School** organized by LIESMARS, Wuhan University. Took courses on **Computer Vision and Digital Image Processing** with the instructor Dr. Peng Fu from Harrisburg University, USA.

01-07 July, 2019 — Participated in **2019 Geo-Informatics Summer School** organized by LIESMARS, Wuhan University and Collaborative Innovation Center of Geospatial Technology. Took courses on **Microwave Remote Sensing** with the instructor Prof. Dr. Timo Balz.

Seminars, Conferences, Workshops, Trainings (continued)

2014 Completed three-month long training program titled **Statistical Computing Certificate Course on MS Excel, SPSS and Programming with R** organized by Department of Statistics, University of Rajshahi, Bangladesh.

Certifications

2026 **Advanced Prompt Engineering Techniques** (LinkedIn)
How to Be an Agent Boss: Leading AI Agents at Work (LinkedIn)
How to Learn a New Topic Quickly (LinkedIn)
Power BI: Dashboards for Beginners (LinkedIn)
Foundations of Geographic Information Systems (GIS) (LinkedIn)
2025 **Microsoft Copilot: The Art of Prompt Writing** (LinkedIn)
Introduction to Career Skills in Software Development (LinkedIn)
Programming Foundations: Beyond the Fundamentals (LinkedIn)
Programming Foundations: Fundamentals (LinkedIn)
Practical Project Management: Gantt Charts, PMI, PMP, APM (Udemy)
2024 **AI Python for Beginners: Basics of AI Python Coding** (DeepLearning.AI)
2022 **Introduction to Python** (DataCamp)
Intermediate Python (DataCamp)
Data Science for Everyone (DataCamp)
Learning Excel: Data Analysis (LinkedIn)
Learning Data Analytics: 1 Foundations (LinkedIn)
Communication Foundations (LinkedIn)
Writing in the Sciences (Coursera)
Good Citation Behavior (Clarivate)
Reviewing in the Sciences (Clarivate)

Technical Skills

Programming & DB	Python, JavaScript, R, SQL, LaTeX
GIS & Cloud	Google Earth Engine (GEE), ArcGIS Desktop, QGIS
AI & ML Models	Machine Learning (Random Forest, XGBoost), Deep Learning Architectures, Neural Networks, Ensemble Learning, Cellular Automata, Advanced Prompt Engineering
Data & Analytics	Power BI, IBM SPSS Statistics, MS Excel, Google Sheets
Field Surveying	RTK, GPS, Total Station, UAVs (Data Acquisition & Processing)

Research Expertise

Remote Sensing & GIS	•Generalized urban impervious surface area mapping, urban land cover classification, dynamic LULC change detection, and multi-sensor data fusion (Simple Layer Stacking). •Image processing and texture analysis (GLCM) utilizing optical, SAR, and AlphaEarth's Satellite Embeddings (64 bands).
Spatial Modeling	•Development of novel spectral indices (GISAI, NSAI); predictive modeling, environmental hazard assessment, and risk mapping (e.g., flood prediction).

Research Expertise (continued)

Research Methodology	•Experimental design, quantitative & qualitative analytics, sampling (10-meter resolution scale), socio-economic surveys, academic citations, and peer review.
Management & Strategy	•Project Management (PMP, APM), Gantt Charts, interdisciplinary collaboration, team leadership, public speaking, and SEO.

Language Proficiency

Bengali	Native speaker.
English	Professional Command. IELTS test score: overall 6.5 (2017).
Dialects	Comprehension of Asian English dialects including Indian English and Chinese English.
Urdu	Conversational proficiency.
Hindi	Conversational proficiency.
Chinese	HSK 1. Learning online for HSK 3 exam.
Arabic	Can read. Taking online course for improve overall comprehension.

Extracurricular Activities

EGSC — **English GeoScience Cafe (EGSC)**, LIESMARS, Wuhan University
- **Editorial Member** (2023–2024): Managed academic content
- **Vice Director** (2017–2018): Organized seminars/conferences
- Coordinated field trips and volunteer programs

SERC — **Space & Environment Research Center** (Co-founder)
- **General Secretary** (2014–2015)
- Organized national conference on disaster management

RUPC — **Rajshahi University Press Club**
- **Elected Treasurer** (2012–2014)
- **Senior Executive Member** (2014–2015)
- **Rajshahi University Correspondent** of Daily Janata (2011–2015)

BNCC — **Bangladesh National Cadet Corps (BNCC)**
- **Cadet Corporal** (2010–2013): Rajshahi University Naval Branch, Khulna Flotilla
- **Contingent Commander** (2013): Led the 'Freedom Day Parade' at the University of Rajshahi
- Completed a **12-day Annual Training Camp** at BNS Shaheed Moazzam, Rangamati
- Completed a **5-day Capsule Training Camp** at Khulna Naval Flotilla, Khulna

RUAA — **Rajshahi University Alumni Association**
- Life Member (2025–Present)

References

Available on Request